AKASHDIP MAHAPATRA

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PROFESSIONAL SUMMARY

Results-driven Site Reliability Engineer with a strong foundation in mechanical systems thinking, specializing in production support, observability, and cloud infrastructure. Proven ability to reduce operational toil through Python automation, manage high-availability CI/CD pipelines, and drive Root Cause Analysis (RCA) in enterprise-scale distributed systems.

CORE COMPETENCIES

- **Security & Observability: (SRE):** Datadog, CloudWatch, Vulnerability Management (CVE patching), SLA/SLO tracking, incident management.
- **DevOps & CI/CD:** GitHub Actions (utilizing IAM Role ARNs via OIDC for passwordless authentication), Jenkins, Terraform (Infrastructure as Code), Advanced Git Version Control (resolving complex remote history mismatches, local vs. remote state reconciliation).
- **Programming & Scripting:** Python (Boto3, Data Structures & Algorithms), Bash, C, SQL, Oracle SQL.
- **Cloud Infrastructure:** AWS (Lambda, ECR, ECS, S3, IAM, VPC, EventBridge, MSK/Kafka, EC2, DynamoDB), & fundamentals of Azure (Azure DevOps) and Google Cloud Platform.
- **Container Orchestration & Cloud:** Docker, AWS ECS, ECR, AWS Serverless (Lambda, EventBridge), MSK/Kafka.
- **Operating Systems & Networking:** Linux (Ubuntu, RHEL), TCP/IP, DNS, HTTP/HTTPS, Nginx, Wireshark, nmap.

PROFESSIONAL EXPERIENCE

1. **Tata Consultancy Services (TCS)** | Onsite *Data Engineer (DevOps & Infrastructure Focus under British Airways)* | Nov 2025 - Present
   - Design (with Team) and maintain automated CI/CD deployment lifecycles utilizing GitHub Actions and Docker, enabling seamless continuous delivery for multi-tiered AWS architectures processing large-scale *200 GB to 500 GB IoT data every day, coming from Source Team (dump in S3)*.
   - Support, troubleshoot, and maintain high-throughput event-driven infrastructure (Amazon MSK/Kafka, EventBridge) ensuring strict environment consistency and system availability.
   - Apply SRE principles to container security by performing comprehensive Root Cause Analysis (RCA) on Docker images and Python dependencies, patching critical CVEs to improve system reliability and minimize potential downtime (0 min for most of the Pipelines & for some maximum 30 min downtime).
   - Engineer and maintain a Python and Boto3-based **automation script** currently adopted by our 30-person DevOps team for universal task validation (including S3 data dumps, deployments, and infrastructure updates). Integrate with **Datadog** and **CloudWatch** to track Lambda invocations and errors, **successfully reducing manual operational toil by 90% through continuous, feedback-driven iterations**.
   - Successfully architected and onboarded the Pathfinder consumer to the Cabin Crew Ground activities, Details, and Schedules APIs, ensuring seamless data retrieval.

2. **Tata Consultancy Services (TCS)** | Remote *Network & DevOps Trainee (Apple Infrastructure Project)* | Oct 2025
   - Troubleshot deep Linux OS and networking issues, analyzing **system calls (strace), inodes**, and network traffic using tcpdump and Wireshark.
   - Deployed and configured Nginx reverse proxies and load balancers, successfully resolving HTTP 403/404 errors and DNS routing failures.
   - Hardened server security by implementing strict file permissions, writing custom Bash scripts for system automation, and configuring network firewalls (UFW, iptables, AWS Security Groups).

SYSTEMS ENGINEERING & AUTOMATION PROJECTS

1. **Cloud Cost Optimization & Observability Investigation**
   - Spearheaded an extensive AWS cost anomaly investigation, utilizing CloudTrail and Datadog to identify operational inefficiencies across Lambda, SNS, and SQS services.
   - Analyzed system behaviors and resource allocation to provide actionable insights for reducing cloud spend and optimizing infrastructure performance.

2. **Lightweight Python Web Server (OSI Layer 4/7):**
    - **Bypassed standard web frameworks to build a raw HTTP server from scratch using basic Python TCP Socket Programming, handling low-level GET/POST request parsing.**

3. **Greenfield Enterprise Cloud Architecture & Provisioning**
    - **Collaborate** with cross-functional development and DevOps teams to architect and provision a completely new cloud infrastructure following the decommissioning of the legacy Mercury 2 (M2) environment.
    - Participate in daily system design Knowledge Transfer (KT) sessions to define scalable networking, security, and deployment standards for the greenfield environment.

4. **Data Pipeline & Release Automation (GSE Events)**
    - Developed and deployed Python-based software solutions utilizing Boto3 to automate complex validation workflows, integrating APIs to eliminate manual processes and accelerate software release cycles.
    - Orchestrated data extraction workflows bridging legacy Oracle SQL databases with AWS S3 cloud storage, ensuring reliable data ingestion for downstream analytics.

5. **Network Security & Packet Forensics:**
    - **Engineered a Python-based wireless penetration testing system to scan 802.11 traffic, capture handshakes, and test access point vulnerabilities.**

PUBLICATIONS & APPLIED RESEARCH (Data Science & Machine Learning)

Modeling and Optimization of Surface Roughness of Electrodeposited Nickel Coating using Taguchi and Bonobo Optimizer | ***Jadavpur University Press (ISBN: 978-81-993635-8-8)***

- **Applied the Bonobo Optimizer (a meta-heuristic optimization algorithm) and Taguchi design of experiments to model, predict, and optimize complex physical processes.**
- **Demonstrated advanced capability in data-driven mathematical modeling, statistical analysis, and implementing optimization algorithms to solve multi-variable engineering challenges.**

EDUCATION

Academy of Technology Engineering College | Hooghly, West Bengal *Bachelor of Technology in Mechanical Engineering* | Sep 2021 - 2025

- Applied mechanical systems-design principles and analytical problem-solving skills to complex cloud infrastructure and software architecture.
- **Mechanical Capabilities & Projects:** AutoCAD, SolidWorks (Advanced CAD, Sheet-Metal, Weldments), Ansys, MATLAB | **Key Focus:** Thermodynamics, Solid/Fluid Mechanics | **Applied Work:** 3D-printed working drone, fully functional V8 engine design, published academic research.

CERTIFICATIONS & QUALIFICATIONS

1. AWS Certified Cloud Practitioner **(Perfect Score: 1000/1000)**
2. Microsoft Certified: Azure Network Engineer Associate
3. Google Cloud Certified Professional
4. Oracle Cloud Infrastructure 2024 Certified Foundations Associate
5. GitHub Certifications: Advanced Security, Actions, and Administration
6. Postman Student Expert
7. NASA Open Science 101 Qualification